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Impact tool

Abstract

Decreased visibility of a workpiece for an operator is reduced. An impact tool includes a motor, a striker rotatable by the motor, an anvil strikable by the striker in a rotation direction, a hammer case accommodating the striker and having a groove, and a light received in the groove. The light illuminates an area adjacent to a front end of the anvil.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Japanese Patent Application No. 2022-192312, filed on Nov. 30, 2022, and Japanese Patent Application No. 2023-173015, filed on Oct. 4, 2023, the entire contents of which are hereby incorporated by reference.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an impact tool.

2. Description of the Background

(3) In the technical field of impact tools, lighting systems for power tools are known as described in U.S. Patent Application Publication No. 2016/0354889. The lighting systems for power tools include one or more chip-on-board light-emitting diodes (COB LEDs) as light units.

BRIEF SUMMARY

(4) COB lights emit light with high intensity. When light emitted from a COB light at least partially reaches the eyes of an operator, such light may cause glare to the operator and decrease the visibility of a workpiece. A COB light may receive a shock when, for example, an impact tool including the COB light is dropped, and may have deteriorating light emission performance.

(5) One or more aspects of the present disclosure are directed to a technique for reducing decreased visibility of a workpiece for an operator, and a technique for protecting a light and reducing deterioration in the emission performance of the light.

(6) A first aspect of the present disclosure provides an impact tool, including: a motor; a striker rotatable by the motor; an anvil strikable by the striker in a rotation direction; a hammer case accommodating the striker, the hammer case having a groove; and a light received in the groove, the light being configured to illuminate an area adjacent to a front end of the anvil.

(7) A second aspect of the present disclosure provides an impact tool, including: a motor; a striker rotatable by the motor; an anvil strikable by the striker in a rotation direction; a light configured to illuminate an area adjacent to a front end of the anvil; and a hammer case accommodating the striker and holding the light, the hammer case including a projection located radially outward from the light.

(8) A third aspect of the present disclosure provides an impact tool, including: a motor; a striker located frontward from the motor, the striker being drivable by the motor; an anvil strikable by the striker in a rotation direction; a hammer case accommodating the striker; a light unit being annular, the light unit being located at a front of the hammer case; and a bumper being annular and comprising an elastomer, the bumper being located outward in a radial direction from the light unit.

(9) The impact tool according to any one of the above aspects of the present disclosure reduces decreased visibility of a workpiece for an operator. The impact tool according to any one of the above aspects of the present disclosure protects the light and reduces deterioration in the emission performance of the light.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of an impact tool according to an embodiment as viewed from the front.

(2) FIG. 2 is a side view of an upper portion of the impact tool according to the embodiment.

(3) FIG. 3 is a longitudinal sectional view of the upper portion of the impact tool according to the embodiment.

(4) FIG. 4 is a horizontal sectional view of the upper portion of the impact tool according to the embodiment.

(5) FIG. 5 is a partial sectional view of a light unit in the embodiment.

(6) FIG. 6 is a partial sectional view of the light unit in the embodiment.

(7) FIG. 7 is an exploded perspective view of the upper portion of the impact tool according to the embodiment as viewed from the front.

(8) FIG. 8 is a perspective view of the light unit in the embodiment as viewed from the front.

(9) FIG. 9 is a perspective view of the light unit in the embodiment as viewed from the rear.

(10) FIG. 10 is an exploded perspective view of the impact tool according to the embodiment as

viewed from the front.

(11) FIG. 11 is a front view of the light unit in the embodiment.

(12) FIG. 12 is a perspective view of the upper portion of the impact tool according to the embodiment as viewed from the front.

(13) FIG. 13 is a perspective view of the upper portion of the impact tool according to the embodiment as viewed from the rear.

(14) FIG. 14 is a side view of the upper portion of the impact tool according to the embodiment.

(15) FIG. 15 is a partial longitudinal sectional view of the impact tool according to the embodiment.

(16) FIG. 16 is a partially enlarged longitudinal sectional view of the impact tool according to the embodiment.

(17) FIG. 17 is a horizontal sectional view of the upper portion of the impact tool according to the embodiment.

(18) FIG. 18 is a horizontal sectional view of the upper portion of the impact tool according to the embodiment.

(19) FIG. 19 is an exploded perspective view of the upper portion of the impact tool according to the embodiment as viewed from the front.

(20) FIG. 20 is an exploded perspective view of a bumper and the light unit in the embodiment as viewed from the rear.

(21) FIG. 21 is an exploded front view of the upper portion of the impact tool according to the embodiment.

(22) FIG. 22 is a partially enlarged longitudinal sectional view of an impact tool according to a modification.

DETAILED DESCRIPTION

(23) A first aspect of the present disclosure provides an impact tool (1), comprising: a motor (6); a hammer (47) rotatable by the motor (6); an anvil (10) strikable by the hammer (47) in a rotation direction; a hammer case (4) accommodating the hammer (47), the hammer case (4) having a groove (4E); and a light (50) received in the groove (4E), the light (50) being configured to illuminate an area adjacent to a front end of the anvil (10).

(24) A second aspect of the present disclosure provides the impact tool (1) according to the first aspect in which the groove (4E) is recessed rearward from a front end of the hammer case (4).

(25) A third aspect of the present disclosure provides the impact tool (1) according to the first aspect or the second aspect in which the groove (4E) surrounds the anvil (10).

(26) A fourth aspect of the present disclosure provides the impact tool (1) according to any one of the first to third aspects, further comprising: an optical member (57) including a light transmitter (57C) located frontward from the light (50), wherein the optical member (57) is at least partially received in the groove (4E).

(27) A fifth aspect of the present disclosure provides the impact tool (1) according to the fourth aspect in which the light transmitter (57C) is received in the groove (4E).

(28) A sixth aspect of the present disclosure provides the impact tool (1) according to the fifth aspect in which the optical member (57) includes an outer cylinder (57A) located radially outward from the light (50), and an inner cylinder (57B) located radially inward from the light (50), and the outer cylinder (57A) and the inner cylinder (57B) are located in the groove (4E).

(29) A seventh aspect of the present disclosure provides an impact tool (1), comprising: a motor (6); a hammer (47) rotatable by the motor (6); an anvil (10) strikable by the hammer (47) in a rotation direction; a light (50) configured to illuminate an area adjacent to a front end of the anvil (10); and a hammer case (4) accommodating the hammer (47) and holding the light (50), the hammer case (4) including a projection (4D) located radially outward from the light (50).

(30) An eighth aspect of the present disclosure provides the impact tool (1) according to the seventh aspect in which the projection (4D) protrudes frontward from the light (50).

(31) A ninth aspect of the present disclosure provides the impact tool (1) according to the seventh aspect or the eighth aspect in which the projection (4D) surrounds the anvil (10).

(32) A tenth aspect of the present disclosure provides the impact tool (1) according to the third aspect or the ninth aspect in which the light (50) includes a substrate (51), and a light-emitting diode chip (52) on a front surface of the substrate (51), and the substrate (51) surrounds the anvil (10).

(33) An eleventh aspect of the present disclosure provides the impact tool (1) according to the ninth aspect, further comprising: an optical member (57) including a light transmitter (57C) located frontward from the light (50), wherein the projection (4D) is located radially outward from the optical member (57).

(34) A twelfth aspect of the present disclosure provides the impact tool (1) according to the eleventh aspect in which the light transmitter (57C) has an emission surface (57F) aligned with or located rearward from a front end face of the projection (4D).

(35) A thirteenth aspect of the present disclosure provides the impact tool (1) according to the fourth aspect or the eleventh aspect in which the anvil (10) includes an anvil shaft (10C) located radially inward from the optical member (57), the hammer case (4) includes a front cylinder (4B) surrounding the anvil shaft (10C), the impact tool (1) further comprises a snap ring (16) received in a snap ring groove (4F) on the front cylinder (4B), and the snap ring (16) supports the optical member (57) from front.

(36) A fourteenth aspect of the present disclosure provides the impact tool (1) according to the thirteenth aspect in which the optical member (57) includes an inner circumferential protrusion (57G) located radially inward from the light transmitter (57C) and frontward from an emission surface (57F) of the light transmitter (57C), and the snap ring (16) supports the inner circumferential protrusion (57G).

(37) A fifteenth aspect of the present disclosure provides the impact tool (1) according to the fourteenth aspect in which the optical member (57) includes a stopper (57H) supporting an end of the snap ring (16).

(38) A sixteenth aspect of the present disclosure provides the impact tool (1) according to the fourteenth aspect, further comprising: a shock absorber (80) between a rear surface of the light (50) and the hammer case (4).

(39) A seventeenth aspect of the present disclosure provides the impact tool (1) according to the fourth aspect or the eleventh aspect in which the optical member (57) includes a protrusion (57D) located downward from the light transmitter (57C), the impact tool (1) further comprises a housing (2) fastened to a rear of the hammer case (4) and accommodating the motor (6), and the housing (2) includes a cover (21C) covering the protrusion (57D) from front.

(40) An eighteenth aspect of the present disclosure provides the impact tool (1) according to the seventeenth aspect in which the hammer case (4) has a facing surface (4J) facing an upper surface (21D) of the cover (21C), and the facing surface (4J) extends in a lateral direction.

(41) A nineteenth aspect of the present disclosure provides the impact tool (1) according to the first aspect or the seventh aspect, further comprising: a housing (2) fastened to a rear of the hammer case (4) with a screw (5), the housing (2) accommodating the motor (6), wherein the hammer case (4) includes a screw boss (4H) engaged with the screw (5), and at least a part of the housing (2) covers the screw boss (4H).

(42) A twentieth aspect of the present disclosure provides the impact tool (1) according to the fourth aspect or the eleventh aspect in which the anvil (10) includes an anvil shaft (10C) located radially inward from the optical member (57), the impact tool (1) further comprises an anvil bearing (46) held by the hammer case (4) and supporting the anvil shaft (10C) in a rotatable manner, and the hammer (47) has a front end located frontward from a rear end of the anvil bearing (46).

(43) A twenty-first aspect of the present disclosure provides the impact tool (1) according to any

one of the first to twentieth aspects in which the light (50) is a chip-on-board light (50).

(44) A twenty-second aspect of the present disclosure provides an impact tool (1B), comprising: a motor (6); a striker (9) located frontward from the motor (6), the striker (9) being drivable by the motor (6); an anvil (10) strikable by the striker (9) in a rotation direction; a hammer case (104) accommodating the striker (9); a light unit (18) being annular, the light unit (18) being located at a front of the hammer case (104); and a bumper (90) being annular and comprising an elastomer, the bumper (90) being located outward in a radial direction from the light unit (18).

(45) A twenty-third aspect of the present disclosure provides the impact tool (1B) according to the twenty-second aspect in which the light unit (18) is aligned with at least a part of the bumper (90) in a front-rear direction.

(46) A twenty-fourth aspect of the present disclosure provides the impact tool (1B) according to the twenty-second aspect in which the hammer case (104) has a groove (4E) being annular on a front portion of the hammer case (104), and the light unit (18) is received in the groove (4E).

(47) A twenty-fifth aspect of the present disclosure provides the impact tool (1B) according to the twenty-fourth aspect, further comprising: an anvil bearing (146) supporting the anvil (10) in a rotatable manner, wherein the hammer case (104) includes a rear cylinder (4A) accommodating the striker (9), a front cylinder (4B) located frontward from the rear cylinder (4A), the front cylinder (4B) holding the anvil bearing (146), an annular portion (4C) connecting a front end of the rear cylinder (4A) and a rear end of the front cylinder (4B), and a projection (4D) protruding frontward from an outer edge of the annular portion (4C), the groove (4E) is between the front cylinder (4B) and the projection (4D), and the bumper (90) covers an outer circumferential surface of the projection (4D).

(48) A twenty-sixth aspect of the present disclosure provides the impact tool (1B) according to the twenty-fifth aspect in which the bumper (90) covers a front end face of the projection (4D).

(49) A twenty-seventh aspect of the present disclosure provides the impact tool (1B) according to the twenty-fifth aspect in which the bumper (90) includes a cylindrical portion (91) covering the outer circumferential surface of the projection (4D), and a protrusion (93) protruding inward in the radial direction from an inner circumferential surface of the cylindrical portion (91), the projection (4D) has a bumper groove (4L) on the outer circumferential surface, and the protrusion (93) is received in the bumper groove (4L).

(50) A twenty-eighth aspect of the present disclosure provides the impact tool (1B) according to the twenty-fifth aspect in which the light unit (18) is aligned with at least a part of the anvil bearing (146) in a front-rear direction.

(51) A twenty-ninth aspect of the present disclosure provides the impact tool (1B) according to the twenty-second aspect in which the bumper (90) has a rubber hardness Hs JIS A of less than or equal to 100.

(52) A thirtieth aspect of the present disclosure provides the impact tool (1B) according to the twenty-second aspect in which the striker (9) includes a hammer (47) rotatable by the motor (6), and the bumper (90) is located frontward from the hammer (47) in a front-rear direction.

(53) One or more embodiments will now be described with reference to the drawings. In the embodiments, the positional relationships between the components will be described using the directional terms such as right and left (or lateral), front and rear (or frontward and rearward), and up and down (or vertical). The terms indicate relative positions or directions with respect to the center of an impact tool.

First Embodiment

(54) Impact Tool

(55) FIG. 1 is a perspective view of an impact tool 1 according to the present embodiment as viewed from the front. FIG. 2 is a side view of an upper portion of the impact tool 1. FIG. 3 is a longitudinal sectional view of the upper portion of the impact tool 1. FIG. 4 is a horizontal sectional view of the upper portion of the impact tool 1.

(56) The impact tool **1** according to the embodiment is a power tool including an electric motor **6** as a power source. A direction parallel to a rotation axis AX of the motor **6** is referred to as an axial direction for convenience. A direction about the rotation axis AX is referred to as a circumferential direction or circumferentially, or a rotation direction for convenience. A direction radial from the rotation axis AX is referred to as a radial direction or radially for convenience. A position nearer the rotation axis AX in the radial direction, or a radial direction toward the rotation axis AX, is referred to as inward in the radial direction or radially inward for convenience. A position farther from the rotation axis AX in the radial direction, or a radial direction away from the rotation axis AX, is referred to as outward in the radial direction or radially outward for convenience. In the embodiments, the rotation axis AX extends in a front-rear direction. A first axial direction is frontward (from the rear to the front), and a second axial direction is rearward (from the front to the rear).

(57) The impact tool **1** according to the embodiment is an impact wrench. The impact tool **1** includes a housing **2**, a rear cover **3**, a hammer case **4**, screws **5**, the motor **6**, a reducer **7**, a spindle **8**, a striker **9**, an anvil **10**, a fan **12**, a battery mount **13**, a trigger lever **14**, a forward-reverse switch lever **15**, a snap ring **16**, a seal **17**, and a light unit **18**.

(58) The housing **2** is formed from a synthetic resin. The housing **2** in the embodiment is formed from nylon. The housing **2** includes a left housing **2L** and a right housing **2R**. The right housing **2R** is located on the right of the left housing **2L**. The left and the right housings **2L** and **2R** are fastened together with multiple screws **2S**. The housing **2** includes a pair of housing halves.

(59) The housing **2** includes a motor compartment **21**, a grip **22**, and a battery holder **23**.

(60) The motor compartment **21** is cylindrical. The motor compartment **21** accommodates the motor **6**, a part of a bearing box **24**, and a rear portion of the hammer case **4**. The motor compartment **21** includes screw bosses **2H**.

(61) The grip **22** protrudes downward from the motor compartment **21**. The trigger lever **14** is located in an upper portion of the grip **22**. The grip **22** is grippable by an operator.

(62) The battery holder **23** is connected to the lower end of the grip **22**. The battery holder **23** has larger outer dimensions than the grip **22** in the front-rear direction and in the lateral direction.

(63) The rear cover **3** is formed from a synthetic resin. The rear cover **3** is located behind the motor compartment **21**. The rear cover **3** accommodates at least a part of the fan **12**. The fan **12** is located radially inward from the rear cover **3**. The rear cover **3** covers an opening in the rear end of the motor compartment **21**. The rear cover **3** is fastened to the rear end of the motor compartment **21** with screws **3S**.

(64) The motor compartment **21** has inlets **19**. The rear cover **3** has outlets **20**. Air outside the housing **2** flows into the internal space of the housing **2** through the inlets **19**. Air inside the housing **2** flows out of the housing **2** through the outlets **20**.

(65) The hammer case **4** serves as a gear case accommodating the reducer **7**. The hammer case **4** accommodates the reducer **7**. The hammer case **4** accommodates the spindle **8**. The hammer case **4** accommodates the striker **9**. The hammer case **4** accommodates a part of the anvil **10**. The hammer case **4** is formed from a metal. The hammer case **4** in the embodiment is formed from aluminum. The hammer case **4** is cylindrical.

(66) The hammer case **4** includes a rear cylinder **4A**, a front cylinder **4B**, an annular portion **4C**, and screw bosses **4H**. The front cylinder **4B** is located frontward from the rear cylinder **4A**. The rear cylinder **4A** has a larger outer diameter than the front cylinder **4B**. The rear cylinder **4A** has a larger inner diameter than the front cylinder **4B**. The annular portion **4C** connects the front end of the rear cylinder **4A** and the rear end of the front cylinder **4B**.

(67) The hammer case **4** is connected to the front of the motor compartment **21**. The motor compartment **21** is fastened to the rear of the hammer case **4** with the screws **5**. The screws **5** are placed into openings of the screw bosses **2H** from the rear, and then into threaded holes of the screw bosses **4H**. The screw bosses **2H** are four screw bosses **2H** arranged circumferentially. The

screw bosses **4H** are four screw bosses **4H** arranged circumferentially. The screws **5** are four screws **5** arranged circumferentially. As shown in FIG. **2**, the screws **2S** overlap the screws **5** in the front-rear direction and in a vertical direction. The screws **2S** are thus unremovable before the screws **5** are removed. The screws **2S** are removable after the screws **5** are removed. The left housing **2L** and the right housing **2R** are fastened with the screws **2S** before the hammer case **4** and the motor compartment **21** are fastened with the screws **5**.

(68) The bearing box **24** is fastened to the rear of the rear cylinder **4A**. The reducer **7** is at least partially located inside the bearing box **24**. The hammer case **4** is held between the left housing **2L** and the right housing **2R**. A part of the bearing box **24** and a rear portion of the rear cylinder **4A** are accommodated in the motor compartment **21**. The bearing box **24** is fixed to the motor compartment **21** and the hammer case **4**.

(69) The motor **6** is a power source for the impact tool **1**. The motor **6** generates a rotational force. The motor **6** is an electric motor. The motor **6** is an inner-rotor brushless motor. The motor **6** includes a stator **26** and a rotor **27**. The stator **26** is supported on the motor compartment **21**. The rotor **27** is at least partially located inside the stator **26**. The rotor **27** rotates relative to the stator **26**. The rotor **27** rotates about the rotation axis **AX** extending in the front-rear direction.

(70) The stator **26** includes a stator core **28**, a front insulator **29**, a rear insulator **30**, and multiple coils **31**.

(71) The stator core **28** is located outward in the radial direction from the rotor **27**. The stator core **28** includes multiple steel plates stacked on one another. The steel plates are metal plates formed from iron as a main component. The stator core **28** is cylindrical. The stator core **28** has multiple teeth to support the coils **31**.

(72) The front insulator **29** is located at the front of the stator core **28**. The rear insulator **30** is located at the rear of the stator core **28**. The front insulator **29** and the rear insulator **30** are electrical insulating members formed from a synthetic resin. The front insulator **29** covers parts of the surfaces of the teeth. The rear insulator **30** covers parts of the surfaces of the teeth.

(73) The coils **31** are attached to the stator core **28** with the front insulator **29** and the rear insulator **30** in between. The coils **31** surround the teeth on the stator core **28** with the front insulator **29** and the rear insulator **30** in between. The coils **31** and the stator core **28** are electrically insulated from each other by the front insulator **29** and the rear insulator **30**. The coils **31** are connected to one another with fusing terminals **38**.

(74) The rotor **27** rotates about the rotation axis **AX**. The rotor **27** includes a rotor core **32**, a rotor shaft **33**, a rotor magnet **34**, and a sensor magnet **35**.

(75) The rotor core **32** and the rotor shaft **33** are formed from steel. In the embodiment, the rotor core **32** and the rotor shaft **33** are integral with each other. The rotor shaft **33** includes a front portion protruding frontward from the front end face of the rotor core **32**. The rotor shaft **33** includes a rear portion protruding rearward from the rear end face of the rotor core **32**.

(76) The rotor magnet **34** is fixed to the rotor core **32**. The rotor magnet **34** is cylindrical. The rotor magnet **34** surrounds the rotor core **32**.

(77) The sensor magnet **35** is fixed to the rotor core **32**. The sensor magnet **35** is annular. The sensor magnet **35** is located on the front end face of the rotor core **32** and the front end face of the rotor magnet **34**.

(78) A sensor board **37** is attached to the front insulator **29**. The sensor board **37** is fastened to the front insulator **29** with a screw **29S**. The sensor board **37** includes an annular circuit board and a magnetic sensor supported on the circuit board. The sensor board **37** at least partially faces the sensor magnet **35**. The magnetic sensor detects the position of the sensor magnet **35** to detect the position of the rotor **27** in the rotation direction.

(79) The rotor shaft **33** includes the rear portion rotatably supported by a rotor bearing **39**. The rotor bearing **39** includes a front portion rotatably supported by a rotor bearing **40**. The rotor bearing **39** is held by the rear cover **3**. The rotor bearing **40** is held by the bearing box **24**. The front

end of the rotor shaft **33** is located in the internal space of the hammer case **4** through an opening in the bearing box **24**.

(80) A pinion gear **41** is located on the front end of the rotor shaft **33**. The pinion gear **41** is connected to at least a part of the reducer **7**. The rotor shaft **33** is connected to the reducer **7** with the pinion gear **41** in between.

(81) The reducer **7** transmits a rotational force from the motor **6** to the spindle **8** and the anvil **10**. The reducer **7** is accommodated in the rear cylinder **4A** in the hammer case **4**. The reducer **7** includes multiple gears. The reducer **7** is located frontward from the motor **6**. The reducer **7** connects the rotor shaft **33** and the spindle **8**. The rotor **27** drives the gears in the reducer **7**. The reducer **7** transmits rotation of the rotor **27** to the spindle **8**. The reducer **7** rotates the spindle **8** at a lower rotational speed than the rotor shaft **33**. The reducer **7** includes a planetary gear assembly.

(82) The reducer **7** includes multiple planetary gears **42** and an internal gear **43**. The multiple planetary gears **42** surround the pinion gear **41**. The internal gear **43** surrounds the multiple planetary gears **42**. The pinion gear **41**, the planetary gears **42**, and the internal gear **43** are accommodated in the hammer case **4** and the bearing box **24**. Each planetary gear **42** meshes with the pinion gear **41**. The planetary gears **42** are rotatably supported by the spindle **8** with a pin **42P**. The spindle **8** is rotated by the planetary gears **42**. The internal gear **43** has internal teeth that mesh with the planetary gears **42**. The internal gear **43** is fixed to the bearing box **24**. The internal gear **43** is constantly nonrotatable relative to the bearing box **24**.

(83) When the rotor shaft **33** rotates as driven by the motor **6**, the pinion gear **41** rotates, and the planetary gears **42** revolve about the pinion gear **41**. The planetary gears **42** revolve while meshing with the internal teeth on the internal gear **43**. The spindle **8**, which is connected to the planetary gears **42** with the pin **42P**, thus rotates at a lower rotational speed than the rotor shaft **33**.

(84) The spindle **8** rotates with a rotational force from the motor **6**. The spindle **8** is located frontward from at least a part of the motor **6**. The spindle **8** is located frontward from the stator **26**. The spindle **8** is at least partially located frontward from the rotor **27**. The spindle **8** is at least partially located in front of the reducer **7**. The spindle **8** is rotated by the rotor **27**. The spindle **8** rotates with a rotational force from the rotor **27** transmitted through the reducer **7**.

(85) The spindle **8** includes a flange **8A** and a spindle shaft **8B**. The spindle shaft **8B** protrudes frontward from the flange **8A**. The planetary gears **42** are rotatably supported by the flange **8A** with the pin **42P**. The rotation axis of the spindle **8** aligns with the rotation axis **AX** of the motor **6**. The spindle **8** rotates about the rotation axis **AX**.

(86) The spindle **8** is rotatably supported by a spindle bearing **44**. The spindle bearing **44** is held by the bearing box **24**. The spindle **8** includes a ring portion **8C** protruding rearward from the rear of the flange **8A**. The spindle bearing **44** is located inward from the ring portion **8C**. The spindle bearing **44** in the embodiment includes an outer ring connected to the ring portion **8C**. The spindle bearing **44** includes an inner ring supported by the bearing box **24**.

(87) The striker **9** is driven by the motor **6**. A rotational force from the motor **6** is transmitted to the striker **9** through the reducer **7** and the spindle **8**. The striker **9** strikes the anvil **10** in the rotation direction with a rotational force of the spindle **8** rotated by the motor **6**. The striker **9** includes a hammer **47**, balls **48**, and coil springs **49**. The striker **9** including the hammer **47** is accommodated in the hammer case **4**.

(88) The hammer **47** is located frontward from the reducer **7**. The hammer **47** is accommodated in the rear cylinder **4A**. The hammer **47** surrounds the spindle shaft **8B**. The hammer **47** is held by the spindle shaft **8B**. The balls **48** are between the spindle shaft **8B** and the hammer **47**. The coil springs **49** are supported by the flange **8A** and the hammer **47**.

(89) The hammer **47** includes an annular body **47D**, a rear outer cylinder **47E**, a front outer cylinder **47F**, an inner cylinder **47G**, a hammer groove **47A**, and two hammer projections **47B**. The rear outer cylinder **47E** protrudes rearward from an outer circumference of the body **47D**. The front outer cylinder **47F** protrudes frontward from the outer circumference of the body **47D**. The inner

cylinder 47G protrudes rearward from an inner circumference of the body 47D.

(90) The body 47D surrounds the spindle shaft 8B. The body 47D is annular. The rear outer cylinder 47E and the inner cylinder 47G both protrude rearward from the body 47D. A recess 47C is defined by the rear surface of the body 47D, the inner circumferential surface of the rear outer cylinder 47E, and the outer circumferential surface of the inner cylinder 47G. The recess 47C is recessed frontward from the rear end of the hammer 47. The recess 47C is ring-shaped. The hammer projections 47B protrude frontward from the body 47D. The hammer projections 47B protrude inward in the radial direction from the inner circumferential surface of the front outer cylinder 47F. The rear outer cylinder 47E and the front outer cylinder 47F allow the hammer 47 to generate a greater inertial force in the rotation direction.

(91) The hammer 47 is rotated by the motor 6. A rotational force from the motor 6 is transmitted to the hammer 47 through the reducer 7 and the spindle 8. The hammer 47 is rotatable together with the spindle 8 with the rotational force of the spindle 8 rotated by the motor 6. The rotation axis of the hammer 47 and the rotation axis of the spindle 8 align with the rotation axis AX of the motor 6. The hammer 47 rotates about the rotation axis AX.

(92) The balls 48 are formed from a metal such as steel. The balls 48 are between the spindle shaft 8B and the hammer 47. The spindle 8 has a spindle groove 8D. The spindle groove 8D receives at least parts of the balls 48. The spindle groove 8D is on the outer circumferential surface of the spindle shaft 8B. The hammer 47 has the hammer groove 47A. The hammer groove 47A receives at least parts of the balls 48. The hammer groove 47A is on the inner surface of the inner cylinder 47G. The balls 48 are between the spindle groove 8D and the hammer groove 47A. The balls 48 roll along the spindle groove 8D and the hammer groove 47A. The hammer 47 is movable together with the balls 48. The spindle 8 and the hammer 47 are movable relative to each other in the axial direction and in the rotation direction within a movable range defined by the spindle groove 8D and the hammer groove 47A.

(93) The coil springs 49 generate an elastic force for moving the hammer 47 forward. The coil springs 49 in the embodiment include a first coil spring 49A and a second coil spring 49B. The first coil spring 49A and the second coil spring 49B are located parallel to each other. The coil springs 49 are between the flange 8A and the hammer 47. The ring-shaped recess 47C is on the rear surface of the hammer 47. The recess 47C is recessed frontward from the rear surface of the hammer 47. A washer 45 is received in the recess 47C. The washer 45 is supported by the body 47D with balls 11 in between. The balls 11 are received in a ball groove 47H on the rear surface of the body 47D. The rear ends of the coil springs 49 are supported by the flange 8A. The front ends of the coil springs 49 are received in the recess 47C and supported by the washer 45.

(94) The anvil 10 is an output unit in the impact tool 1 and is operable with a rotational force from the motor 6. The anvil 10 rotates with the rotational force from the motor 6. The anvil 10 is at least partially located frontward from the hammer 47.

(95) The anvil 10 includes a rod-like anvil shaft 10C and anvil projections 10D. The anvil shaft 10C has a substantially rectangular outer shape as viewed in a direction perpendicular to the rotation axis AX. The anvil shaft 10C receives a socket that is a tip tool. The anvil 10 has a recess 10B at its rear end. The spindle shaft 8B includes a protrusion 8E on its front end. The recess 10B on the rear end of the anvil 10 receives the protrusion 8E on the front end of the spindle shaft 8B. The anvil projections 10D are located on the rear end of the anvil 10. The anvil projections 10D protrude outward in the radial direction from the rear end of the anvil shaft 10C. The protrusion 8E has two grease grooves 8F.

(96) The anvil 10 is rotatably supported by an anvil bearing 46. The rotation axis of the anvil 10, the rotation axis of the hammer 47, and the rotation axis of the spindle 8 align with the rotation axis AX of the motor 6. The anvil 10 rotates about the rotation axis AX. The anvil bearing 46 is located inward from the front cylinder 4B. The anvil bearing 46 is held by the front cylinder 4B in the hammer case 4. The front cylinder 4B surrounds the anvil shaft 10C. The anvil bearing 46 supports

the anvil shaft **10C** in a rotatable manner. The anvil bearing **46** in the embodiment includes an outer sleeve **46A** and an inner sleeve **46B**. The inner sleeve **46B** is located radially inward from the outer sleeve **46A**.

(97) The hammer projections **47B** can come in contact with the anvil projections **10D**. When the motor **6** operates with the hammer projections **47B** and the anvil projections **10D** in contact with each other, the anvil **10** rotates together with the hammer **47** and the spindle **8**.

(98) The anvil **10** is strikable by the hammer **47** in the rotation direction. When, for example, the anvil **10** receives a higher load in a screwing operation, the anvil **10** cannot rotate with power generated by the motor **6** alone. This stops the rotation of the anvil **10** and the hammer **47**. The spindle **8** and the hammer **47** are movable relative to each other in the axial direction and in the circumferential direction with the balls **48** in between. When the hammer **47** stops rotating, the spindle **8** continues to rotate with power generated by the motor **6**. When the hammer **47** stops rotating and the spindle **8** rotates, the balls **48** move backward as being guided along the spindle groove **8D** and the hammer groove **47A**. The hammer **47** receives a force from the balls **48** to move backward with the balls **48**. In other words, the hammer **47** moves backward when the anvil **10** stops rotating and the spindle **8** rotates. As the hammer **47** moves backward, the hammer projections **47B** come out of contact with the anvil projections **10D**.

(99) The coil springs **49** generate an elastic force for moving the hammer **47** forward. The hammer **47** moved backward moves forward under the elastic force from the coil springs **49**. When moving forward, the hammer **47** receives a force in the rotation direction from the balls **48**. In other words, the hammer **47** moves forward while rotating. The hammer projections **47B** then come in contact with the anvil projections **10D** while rotating. Thus, the anvil projections **10D** are struck by the hammer projections **47B** in the rotation direction. The anvil **10** receives power from the motor **6** and an inertial force from the hammer **47**. The anvil **10** thus rotates about the rotation axis **AX** at high torque.

(100) The fan **12** rotates with a rotational force from the motor **6**. The fan **12** is located rearward from the stator **26**. The fan **12** generates an airflow for cooling the motor **6**. The fan **12** is fastened to at least a part of the rotor **27**. The fan **12** is fastened to the rear of the rotor shaft **33** with a bush **12A**. The fan **12** is between the rotor bearing **39** and the stator **26**. The fan **12** rotates as the rotor **27** rotates. As the rotor shaft **33** rotates, the fan **12** rotates together with the rotor shaft **33**. Thus, air outside the housing **2** flows into the internal space of the housing **2** through the inlets **19** to cool the motor **6**. As the fan **12** rotates, the air passing through the housing **2** flows out of the housing **2** through the outlets **20**.

(101) The battery mount **13** is located in a lower portion of the battery holder **23**. A battery pack **25** is attached to the battery mount **13** in a detachable manner. The battery pack **25** serves as a power supply for the impact tool **1**. The battery pack **25** includes a secondary battery. The battery pack **25** in the embodiment includes a rechargeable lithium-ion battery. The battery pack **25** is attached to the battery mount **13** to power the impact tool **1**. The motor **6** and the light unit **18** are each driven by power supplied from the battery pack **25**.

(102) The trigger lever **14** is located on the grip **22**. The trigger lever **14** is operable by the operator to activate the motor **6**. The trigger lever **14** is operable to switch the motor **6** between the driving state and the stopped state.

(103) The forward-reverse switch lever **15** is located above the grip **22**. The forward-reverse switch lever **15** is operable by the operator. The forward-reverse switch lever **15** is operable to switch the rotation direction of the motor **6** between forward and reverse. This operation switches the rotation direction of the spindle **8**.

(104) Light Unit

(105) FIGS. **5** and **6** are each a partial sectional view of the light unit **18** in the embodiment. FIG. **5** is a cross-sectional view of an upper portion of the light unit **18**. FIG. **6** is a cross-sectional view of a lower portion of the light unit **18**. FIG. **7** is an exploded perspective view of the upper portion of

the impact tool **1** as viewed from the front. FIG. **8** is a perspective view of the light unit **18** as viewed from the front. FIG. **9** is a perspective view of the light unit **18** as viewed from the rear. FIG. **10** is an exploded perspective view of the impact tool **1** as viewed from the front. FIG. **11** is a front view of the light unit **18**.

(106) The light unit **18** emits illumination light. The light unit **18** illuminates the anvil **10** and an area adjacent to the anvil **10** with illumination light. The light unit **18** illuminates the front end of the anvil **10** and an area adjacent to the front end of the anvil **10** with illumination light.

(107) The light unit **18** is located at the front of the hammer case **4**. The light unit **18** surrounds the front cylinder **4B**. The light unit **18** surrounds the anvil shaft **10C** with the front cylinder **4B** in between.

(108) The light unit **18** includes a light **50**. The light **50** in the embodiment includes a chip-on-board light-emitting diode (COB LED). The light **50** is hereafter referred to as a COB light **50** as appropriate. The COB light **50** illuminates the front end of the anvil **10** and the area adjacent to the front end of the anvil **10**.

(109) The COB light **50** includes a substrate **51**, LED chips **52** being light emitters, banks **54**, and a phosphor **55**. The substrate **51** is, for example, an aluminum substrate, a glass fabric base epoxy resin substrate (flame retardant 4 or FR-4 substrate), or a composite base epoxy resin substrate (composite epoxy material 3 or CEM-3 substrate). The LED chips **52** are mounted on the front surface of the substrate **51**. The LED chips **52** are connected to the substrate **51** with gold wires (not shown). The gold wires interconnect the multiple LED chips **52**. The banks **54** are located on the front surface of the substrate **51**. The banks **54** surround the LED chips **52**. One bank **54** is located inward in the radial direction from the LED chips **52**, and the other bank **54** is located outward in the radial direction from the LED chips **52**. The banks **54** define the space for the phosphor **55**. The phosphor **55** covers the LED chips **52** between the banks **54**. A pair of electrodes (not shown) are located outside the banks **54** on the front surface of the substrate **51**. The electrodes may be located on the back surface (rear surface) of the substrate **51**. The pair of electrodes are a positive electrode and a negative electrode. Power output from the battery pack **25** is supplied to the electrodes. The power supplied to the electrodes is supplied to the LED chips **52** through the substrate **51** and the gold wires. The LED chips **52** emit light with power supplied from the battery pack **25**. The voltage of the battery pack **25** is decreased to 5 V by a controller (not shown) and applied to the LED chips **52**. The controller is accommodated in the battery holder **23**. The light unit **18** and the controller are connected with lead wires **180**.

(110) The light unit **18** includes the COB light **50** and an optical member **57**. The COB LED **50** includes the substrate **51**, the multiple LED chips **52**, the banks **54**, and the phosphor **55**.

(111) The substrate **51** is annular. The substrate **51** surrounds the anvil shaft **10C** with the front cylinder **4B** in between. The substrate **51** includes a ring portion **51A** and a support **51B**. The support **51B** protrudes downward from a lower portion of the ring portion **51A**. The substrate **51** surrounds the anvil shaft **10C**.

(112) The LED chips **52** are mounted on the front surface of the ring portion **51A** of the substrate **51**. The LED chips **52** at least partially surround the anvil shaft **10C** with the front cylinder **4B** in between. The multiple LED chips **52** (24 LED chips **52** in the present embodiment) are arranged at intervals in the circumferential direction of the ring portion **51A**.

(113) The banks **54** are located on the front surface of the ring portion **51A** of the substrate **51**. The banks **54** protrude frontward from the front surface of the ring portion **51A**. The banks **54** define the space for the phosphor **55**. The banks **54** are annular. The banks **54** in the embodiment have a double annular structure. More specifically, the banks **54** in the embodiment include an annular first bank **54** and an annular second bank **54**. The first bank **54** is located on the front surface of the ring portion **51A**. The second bank **54** is located outward in the radial direction from the first bank **54** on the front surface of the ring portion **51A**. The first bank **54** is located inward in the radial direction from the LED chips **52**. The second bank **54** is located outward in the radial direction

from the LED chips 52. The LED chips 52 are between the first bank 54 and the second bank 54.

(114) The phosphor 55 is located on the front surface of the ring portion 51A of the substrate 51. The phosphor 55 is annular. The phosphor 55 covers the LED chips 52 between the first bank 54 and the second bank 54.

(115) A pair of lead wires (not shown) are connected to the substrate 51. The lead wires are connected to the electrodes described above. The pair of lead wires are supported on the rear surface of the support 51B. The lead wires may be supported on the front surface of the support 51B.

(116) A current output from the battery pack 25 is supplied to the electrodes through the controller (not shown) and the lead wires 180. The voltage of the battery pack 25 is decreased by the controller (not shown) and applied to the electrodes. The current supplied to the electrodes is supplied to the LED chips 52 through the substrate 51 and the gold wires. The LED chips 52 emit light with the current supplied from the battery pack 25.

(117) The optical member 57 is connected to the COB light 50. The optical member 57 is fixed to the substrate 51. The optical member 57 is formed from a polycarbonate resin. The optical member 57 in the embodiment is formed from a polycarbonate resin containing a white diffusion material. The optical member 57 is milk white. The optical member 57 has a light transmittance of 40 to 70% inclusive. With the milk white optical member 57, the profiles of the LED chips 52 are not easily visible from outside the impact tool 1. The impact tool 1 with this structure has an improved design.

(118) The optical member 57 is at least partially located frontward from the COB light 50. The optical member 57 includes an outer cylinder 57A, an inner cylinder 57B, a light transmitter 57C, a protrusion 57D, an inner circumferential protrusion 57G, stoppers 57H, and a recess 57J.

(119) The outer cylinder 57A is located outward in the radial direction from the inner cylinder 57B. The outer cylinder 57A is located radially outward from the COB light 50. The outer cylinder 57A is located outward in the radial direction from the LED chips 52. The COB light 50 is between the outer cylinder 57A and the inner cylinder 57B in the radial direction. The outer cylinder 57A is located outward in the radial direction from the ring portion 51A of the substrate 51. The inner cylinder 57B is located radially inward from the COB light 50. The inner cylinder 57B is located inward in the radial direction from the ring portion 51A of the substrate 51. The inner cylinder 57B is located inward in the radial direction from the LED chips 52.

(120) The light transmitter 57C is located frontward from the COB light 50. The light transmitter 57C is annular. The light transmitter 57C is located frontward from the LED chips 52. The light transmitter 57C connects the front end of the outer cylinder 57A and the front end of the inner cylinder 57B. The light transmitter 57C faces the front surface of the ring portion 51A. The light transmitter 57C faces the LED chips 52. Light emitted from the LED chips 52 passes through the light transmitter 57C and illuminates an area ahead of the light unit 18.

(121) The light transmitter 57C has an incident surface 57E and an emission surface 57F. Light from the LED chips 52 enters the incident surface 57E. The light passing through the light transmitter 57C exits through the emission surface 57F. The front surface of the ring portion 51A faces the incident surface 57E. The incident surface 57E faces the LED chips 52. The incident surface 57E faces substantially rearward. The emission surface 57F faces substantially frontward.

(122) The protrusion 57D is located downward from the light transmitter 57C. The protrusion 57D protrudes downward from a lower portion of the outer cylinder 57A. The protrusion 57D defines an accommodation space in an upper portion of its rear surface. The support 51B in the substrate 51 is received in the accommodation space defined in the upper portion of the rear surface of the protrusion 57D. The protrusion 57D supports the lead wires 180 extending from the substrate 51.

(123) The inner circumferential protrusion 57G is located radially inward from the light transmitter 57C. The inner circumferential protrusion 57G protrudes frontward from the inner circumference of the light transmitter 57C. The inner circumferential protrusion 57G is located frontward from the

emission surface 57F of the light transmitter 57C. The inner circumferential protrusion 57G is annular.

(124) The stoppers 57H protrude frontward from a lower portion of the inner circumferential protrusion 57G. The stoppers 57H are two stoppers 57H located in the lower portion of the inner circumferential protrusion 57G. The two stoppers 57H are arranged at a distance from each other in the lateral direction. The support 51B in the substrate 51 is received in the recess 57J on the rear portion of the optical member 57. The support 51B fitted into the recess 57J positions the substrate 51 relative to the optical member 57, restricting relative rotation between the substrate 51 and the optical member 57.

(125) The substrate 51 has its rear surface located frontward from the rear end of the outer cylinder 57A and the rear end of the inner cylinder 57B. The substrate 51 has its rear surface fixed to at least a part of the inner circumferential surface of the outer cylinder 57A with an adhesive. The substrate 51 has its rear surface fixed to at least a part of the outer circumferential surface of the inner cylinder 57B with an adhesive. The COB light 50 is fixed to the optical member 57.

(126) The hammer case 4 includes a projection 4D located radially outward from the COB light 50. The projection 4D protrudes frontward from the COB light 50. The projection 4D is substantially annular. The projection 4D surrounds the anvil shaft 10C. The projection 4D is located radially outward from the optical member 57. The emission surface 57F is aligned with the front end face of the projection 4D in the front-rear direction. In other words, the emission surface 57F is flush with the front end face of the projection 4D. The emission surface 57F may be located rearward from the front end face of the projection 4D in the front-rear direction.

(127) The hammer case 4 has a groove 4E radially inward from the projection 4D. The groove 4E is recessed rearward from the front end of the hammer case 4. The groove 4E is between the front cylinder 4B and the projection 4D in the radial direction. The groove 4E surrounds the anvil shaft 10C.

(128) The COB light 50 is received in the groove 4E. The optical member 57 is at least partially received in the groove 4E. The light transmitter 57C is received in the groove 4E. The outer cylinder 57A and the inner cylinder 57B are received in the groove 4E. The inner circumferential protrusion 57G is located frontward from the front end of the groove 4E.

(129) The snap ring 16 is received in a snap ring groove 4F on the front cylinder 4B. The snap ring groove 4F is located frontward from the front end face of the projection 4D and the emission surface 57F of the light transmitter 57C. The snap ring 16 supports the optical member 57 from the front. The snap ring 16 in the embodiment supports the inner circumferential protrusion 57G from the front.

(130) The stoppers 57H support the ends of the snap ring 16. The snap ring 16 has two ends. The snap ring 16 has its left end supported by a left stopper 57H of the two stoppers 57H and its right end supported by a right stopper 57H of the two stoppers 57H. The snap ring 16 has its left end in contact with the left surface of the left stopper 57H and its right end in contact with the right surface of the right stopper 57H. The stoppers 57H restrict rotation of the snap ring 16 relative to the optical member 57.

(131) A shock absorber 80 is located behind the COB light 50. The shock absorber 80 in the embodiment is a sponge ring. The shock absorber 80 is between the rear surface of the COB light 50 and the hammer case 4. The shock absorber 80 has its rear surface in contact with the front surface of the annular portion 4C in the hammer case 4 and its front surface in contact with the rear surface of the substrate 51 in the COB light 50. The shock absorber 80 is compressed between the rear surface of the COB light 50 and the hammer case 4. As described above, the substrate 51 and the optical member 57 are fixed to each other with an adhesive. The snap ring 16 supports the optical member 57 and the COB light 50 from the front, compressing the shock absorber 80. The optical member 57 and the COB light 50 are held between the shock absorber 80 and the snap ring 16 in the front-rear direction. The optical member 57 and the hammer case 4 in the embodiment are

apart from each other. The COB light **50** and the hammer case **4** are apart from each other. The COB light **50** and the optical member **57** are held by the hammer case **4** between the shock absorber **80** and the snap ring **16**.

(132) Hammer Case and Housing

(133) The anvil shaft **10C** is located radially inward from the optical member **57**. The anvil bearing **46** held by the hammer case **4** supports the anvil shaft **10C** in a rotatable manner. The anvil bearing **46** is insert-molded in the hammer case **4**. In the embodiment, the front outer cylinder **47F** in the hammer **47** has its front end located frontward from the rear end of the anvil bearing **46**. More specifically, the hammer **47** at least partially overlaps the anvil bearing **46** in the front-rear direction. The seal **17** is retained by the front cylinder **4B**.

(134) As shown in FIG. 5, the hammer case **4** includes a protrusion **4K**. The protrusion **4K** protrudes rearward from the rear surface of the annular portion **4C**. The outer sleeve **46A** includes a flange **46C**. The flange **46C** extends outward in the radial direction from the rear end of the outer sleeve **46A**. The front outer cylinder **47F** has its front end located frontward from the rear end of the protrusion **4K**. The front outer cylinder **47F** has its front end located frontward from the rear end of the flange **46C**. More specifically, the front end of the front outer cylinder **47F** overlaps the protrusion **4K** and the flange **46C** in the front-rear direction. This increases an inertial force from the hammer **47**. The flange **46C** in contact with the anvil projections **10D** reduces the likelihood of the protrusion **4K** being shaved by the anvil projections **10D**. The snap ring **16** and the seal **17** overlap each other in the front-rear direction. With the seal **17** located rearward, the impact tool **1** is less likely to be elongated in the front-rear direction.

(135) The motor compartment **21** covers lower two screw bosses **4H** of the four screw bosses **4H**. The housing **2** includes a cover **21C** covering the protrusion **57D** on the optical member **57** from the front. At least a part of the light emitted from the COB light **50** and passing through the protrusion **57D** is blocked by the cover **21C**. In other words, the cover **21C** reduces leakage of light from the protrusion **57D**.

(136) As shown in FIG. 11, the hammer case **4** has a facing surface **4J**. The facing surface **4J** faces the upper surface **21D** of the cover **21C**. The facing surface **4J** includes a left facing surface **4J** and a right facing surface **4J**. The left facing surface **4J** faces the upper surface **21D** of the cover **21C** in the left housing **2L**. The right facing surface **4J** faces the upper surface **21D** of the cover **21C** in the right housing **2R**. The facing surface **4J** extends in the lateral direction in a plane perpendicular to the rotation axis **AX**. The facing surface **4J** extends linearly in the plane perpendicular to the rotation axis **AX**. As shown in FIG. 10, the housing **2** includes the left housing **2L** and the right housing **2R**. The left housing **2L** and the right housing **2R** to be joined together can be aligned with each other with the upper surface **21D** and the facing surface **4J** in contact with each other. The left housing **2L** and the right housing **2R** can thus be fixed to each other smoothly using the facing surface **4J**.

Method of Use

(137) The operator operates the trigger lever **14** to activate the motor **6** and cause the LED chips **52** in the COB light **50** to emit light. The COB light **50** emits light with high intensity and thus can brightly illuminate a workpiece.

(138) When light emitted from the LED chips **52** at least partially passes through the outer cylinder **57A**, such light emitted through the outer circumferential surface of the outer cylinder **57A** may reach the eyes of the operator and cause glare to the operator. This may decrease the visibility of the workpiece for the operator. In the embodiment, the light emitted through the outer circumferential surface of the outer cylinder **57A** is blocked by the projection **4D**. This reduces glare to the operator.

(139) The COB light **50** is protected by the projection **4D** on the hammer case **4** when, for example, the impact tool **1** is dropped. This reduces breakage of the COB light **50** and deterioration in the emission performance of the COB light **50**.

(140) As described above, the impact tool **1** according to the embodiment includes the motor **6**, the hammer **47** rotatable by the motor **6**, the anvil **10** strikable by the hammer **47** in the rotation direction, the hammer case **4** accommodating the hammer **47**, and the COB light **50** that illuminates the front end of the anvil **10** and the area adjacent to the front end of the anvil **10**. The hammer case **4** has the groove **4E**. The COB light **50** is received in the groove **4E**.

(141) In the above structure, with the COB light **50** received in the groove **4E**, light emitted frontward from the COB light **50** illuminates a workpiece whereas light emitted radially outward from the COB light **50** is blocked by the hammer case **4**. This reduces the likelihood of light emitted from the COB light **50** entering the eyes of the operator and thus reduces glare to the operator. This structure thus reduces decreased visibility of a workpiece for the operator.

(142) The groove **4E** in the embodiment is recessed rearward from the front end of the hammer case **4**.

(143) This allows light emitted frontward from the COB light **50** to illuminate a workpiece in front of the hammer case **4**.

(144) The groove **4E** in the embodiment surrounds the anvil **10**.

(145) The groove **4E** is thus annular.

(146) The impact tool **1** according to the embodiment includes the optical member **57** including the light transmitter **57C** located frontward from the COB light **50**. The optical member **57** is at least partially received in the groove **4E**.

(147) This allows light emitted frontward from the COB light **50** to illuminate a workpiece in front of the hammer case **4** through the light transmitter **57C** in the optical member **57**. With the optical member **57** received in the groove **4E**, light emitted frontward from the optical member **57** illuminates the workpiece whereas light emitted radially outward from the optical member **57** is blocked by the hammer case **4**.

(148) The light transmitter **57C** in the embodiment is received in the groove **4E**.

(149) With the light transmitter **57C** received in the groove **4E**, light emitted frontward from the optical member **57** illuminates the workpiece whereas light emitted radially outward from the optical member **57** is blocked by the hammer case **4**.

(150) The optical member **57** in the embodiment includes the outer cylinder **57A** located radially outward from the COB light **50** and the inner cylinder **57B** located radially inward from the COB light **50**. The outer cylinder **57A** and the inner cylinder **57B** are received in the groove **4E**.

(151) The optical member **57** thus surrounds the COB light **50**.

(152) The impact tool **1** according to the embodiment includes the motor **6**, the hammer **47** rotatable by the motor **6**, the anvil **10** strikable by the hammer **47** in the rotation direction, the COB light **50** that illuminates an area adjacent to the distal end of the anvil **10**, and the hammer case **4** accommodating the hammer **47**. The hammer case **4** holds the COB light **50**. The hammer case **4** includes the projection **4D** located radially outward from the COB light **50**.

(153) In the above structure, the COB light **50** is protected by the projection **4D** on the hammer case **4** when, for example, the impact tool **1** is dropped. This reduces breakage of the COB light **50** and deterioration in the emission performance of the COB light **50**.

(154) The projection **4D** in the embodiment protrudes frontward from the COB light **50**. The COB light **50** is thus protected sufficiently by the projection **4D**.

(155) The projection **4D** in the embodiment surrounds the anvil **10**.

(156) The projection **4D** is thus annular.

(157) The COB light **50** in the embodiment includes the substrate **51** and the LED chips **52** mounted on the front surface of the substrate **51**. The substrate **51** surrounds the anvil **10**.

(158) The impact tool **1** thus includes the annular COB light **50**.

(159) The impact tool **1** according to the embodiment includes the optical member **57** including the light transmitter **57C** located frontward from the COB light **50**. The projection **4D** is located radially outward from the optical member **57**.

(160) The COB light **50** and the optical member **57** are thus protected by the projection **4D**.

(161) The emission surface **57F** of the light transmitter **57C** in the embodiment is aligned with or located rearward from the front end face of the projection **4D**.

(162) The optical member **57** is thus protected sufficiently by the projection **4D**.

(163) The anvil **10** in the embodiment includes the anvil shaft **10C** located radially inward from the optical member **57**. The hammer case **4** includes the front cylinder **4B** surrounding the anvil shaft **10C**. The impact tool **1** includes the snap ring **16** received in the snap ring groove **4F** on the front cylinder **4B**. The snap ring **16** supports the optical member **57** from the front.

(164) This reduces the likelihood of the optical member **57** and the COB light **50** slipping frontward from the hammer case **4**. The snap ring **16** serves as a retainer that reduces slipping of the optical member **57** and the COB light **50** frontward.

(165) The optical member **57** in the embodiment includes the inner circumferential protrusion **57G** located radially inward from the light transmitter **57C** and frontward from the emission surface **57F** of the light transmitter **57C**. The snap ring **16** supports the inner circumferential protrusion **57G**.

(166) The snap ring **16** received in the snap ring groove **4F** stably supports the optical member **57** from the front.

(167) The optical member **57** in the embodiment may include the stoppers **57H** supporting the ends of the snap ring **16**.

(168) This restricts rotation of the snap ring **16** relative to the hammer case **4** and the optical member **57**. The stoppers **57H** serve as rotation locking members that restrict rotation of the snap ring **16**.

(169) The impact tool **1** according to the embodiment includes the shock absorber **80** between the rear surface of the COB light **50** and the hammer case **4**.

(170) This reduces shock to the COB light **50** when, for example, the impact tool **1** is dropped. The COB light **50** held between the shock absorber **80** and the snap ring **16** in the front-rear direction restricts displacement of the COB light **50** and the hammer case **4** relative to each other.

(171) The optical member **57** in the embodiment includes the protrusion **57D** located downward from the light transmitter **57C**. The impact tool **1** includes the housing **2** fastened to the rear of the hammer case **4** and accommodating the motor **6**. The housing **2** includes the cover **21C** covering the protrusion **57D** from the front.

(172) This allows the lead wires connected to the COB light **50** to be supported by the protrusion **57D**. The cover **21C** protects the protrusion **57D**. At least a part of light emitted from the COB light **50** and passing through the protrusion **57D** is blocked by the cover **21C**.

(173) The hammer case **4** in the embodiment has the facing surface **4J** facing the upper surface **21D** of the cover **21C**. The facing surface **4J** extends laterally.

(174) When the housing **2** includes a pair of housing halves including the left housing **2L** and the right housing **2R**, the left housing **2L** and the right housing **2R** can be smoothly fixed to each other using the facing surface **4J**.

(175) The impact tool **1** according to the embodiment includes the housing **2** fastened to the rear of the hammer case **4** with the screws **5** and accommodating the motor **6**. The hammer case **4** includes the screw bosses **4H** engaged with the screws **5**. At least a part of the housing **2** covers the screw bosses **4H**.

(176) The housing **2** thus protects the screw bosses **4H** in the hammer case **4**. The hand of the operator holding the grip **22** in the housing **2** is less likely to touch the hammer case **4** directly.

(177) The anvil **10** in the embodiment includes the anvil shaft **10C** located radially inward from the optical member **57**. The impact tool **1** includes the anvil bearing **46** held by the hammer case **4** and supporting the anvil shaft **10C** in a rotatable manner. The hammer **47** has its front end located frontward from the rear end of the anvil bearing **46**.

(178) With the hammer case **4** and the anvil bearing **46** overlapping each other, the impact tool **1** has less size increase. The impact tool **1** has a smaller dimension in the front-rear direction in its

upper portion (or has a smaller total length).

Second Embodiment

(179) A second embodiment will be described. The same or corresponding components as those in the above embodiment are given the same reference numerals herein and will be described briefly or will not be described.

(180) FIG. **12** is a perspective view of an upper portion of an impact tool **1B** according to the present embodiment as viewed from the front. FIG. **13** is a perspective view of the upper portion of the impact tool **1B** as viewed from the rear. FIG. **14** is a side view of the upper portion of the impact tool **1B**. FIG. **15** is a partial longitudinal sectional view of the impact tool **1B**. FIG. **16** is a partially enlarged longitudinal sectional view of the impact tool **1B**. FIG. **17** is a horizontal sectional view of the upper portion of the impact tool **1B**, taken along line A-A in FIG. **14** as viewed in the direction indicated by the arrows. FIG. **18** is a horizontal sectional view of the upper portion of the impact tool **1B**, taken along line B-B in FIG. **14** as viewed in the direction indicated by the arrows. FIG. **19** is an exploded perspective view of the upper portion of the impact tool **1B** as viewed from the front. FIG. **20** is an exploded perspective view of a bumper **90** and the light unit **18** as viewed from the rear. FIG. **21** is an exploded front view of the upper portion of the impact tool **1B**.

(181) As in the above embodiment, the impact tool **1B** is an impact wrench. The impact tool **1B** includes the housing **2**, the rear cover **3**, a hammer case **104**, the screws **5**, the spindle **8**, the striker **9**, the anvil **10**, the trigger lever **14**, the forward-reverse switch lever **15**, a circlip **116**, the seal **17**, the light unit **18**, and the bumper **90**.

(182) The hammer case **104** accommodates the spindle **8**. The hammer case **104** accommodates the striker **9**. As in the above embodiment, the striker **9** is located frontward from the motor **6**. The striker **9** is driven by the motor **6**. The hammer case **104** accommodates a part of the anvil **10**. The hammer case **104** is formed from a metal. The hammer case **104** in the embodiment is formed from aluminum. The hammer case **104** is cylindrical.

(183) The hammer case **104** includes the rear cylinder **4A**, the front cylinder **4B**, the annular portion **4C**, the projection **4D**, the groove **4E**, the screw bosses **4H**, a bumper groove **4L**, and a circlip groove **4M**. The striker **9** is accommodated in the rear cylinder **4A**. The front cylinder **4B** is located frontward from the rear cylinder **4A**. The rear cylinder **4A** has a larger outer diameter than the front cylinder **4B**. The rear cylinder **4A** has a larger inner diameter than the front cylinder **4B**. The annular portion **4C** connects the front end of the rear cylinder **4A** and the rear end of the front cylinder **4B**. The projection **4D** protrudes frontward from the outer edge of the annular portion **4C**. The projection **4D** is annular in a plane perpendicular to the rotation axis **AX**.

(184) The hammer case **104** is connected to the front of the motor compartment **21** in the housing **2**. The motor compartment **21** accommodates the rear portion of the hammer case **104**. The motor compartment **21** is fastened to the rear of the hammer case **104** with the screws **5**. The screws **5** are placed into openings of the screw bosses **2H** from the rear, and then into threaded holes of the screw bosses **4H**. The screw bosses **2H** are four screw bosses **2H** arranged circumferentially. The screw bosses **4H** are four screw bosses **4H** arranged circumferentially. The screws **5** are four screws **5** arranged circumferentially.

(185) The rear cover **3** is fastened to the motor compartment **21** with the screws **3S** tightened from the rear of the rear cover **3**. The motor compartment **21** is fastened to the hammer case **104** with the screws **5** tightened from the rear of the screw bosses **2H**. A total of six screws (two screws **3S** and four screws **5**) are tightened from the rear, facilitating the assembly of the impact tool **1B**. An assembler of the impact tool **1B** can tighten all the six screws from the rear without changing the orientation of the impact tool **1B**.

(186) The screw bosses **2H** are each located in the upper left, lower left, upper right, and lower right of the rotation axis **AX**. The screw bosses **4H** are each located in the upper left, lower left, upper right, and lower right of the rotation axis **AX**. The screw bosses **4H** include upper left and

upper right screw bosses **4Ha** and lower left and lower right screw bosses **4Hb**. The motor compartment **21** has its front portion covering at least a part of the surface of the hammer case **104**. The motor compartment **21** has its front portion covering at least the surfaces of the lower left and lower right screw bosses **4Hb**. The motor compartment **21** covering the surface of the hammer case **104** has an upper end **2A** located upward from the middle of the hammer case **104** in the vertical direction. The motor compartment **21** covering the surface of the hammer case **104** has a front end **2B** located frontward from the rear end of the bumper **90** in the front-rear direction. In other words, the motor compartment **21** covers a part of the bumper **90**. The forward-reverse switch lever **15** is located in an opening **2C** in a lower portion of the motor compartment **21**.

(187) As shown in FIGS. **12**, **13**, **14**, **17**, and **18**, the motor compartment **21** does not cover the surfaces of the upper left and upper right screw bosses **4Ha**. The motor compartment **21** may cover the surfaces of the upper left and upper right screw bosses **4Ha**.

(188) The hammer case **104** may be heated by, for example, rotation of the spindle **8** and the operation of the striker **9**. The motor compartment **21** covers the surface of the lower portion of the hammer case **104** including the surfaces of the lower left and lower right screw bosses **4Hb**. This reduces the likelihood that the hand of the operator operating the forward-reverse switch lever **15** or the trigger lever **14** directly touches the hammer case **104**.

(189) The spindle **8** rotates with a rotational force from the motor **6**. The striker **9** strikes the anvil **10** in the rotation direction with a rotational force from the spindle **8**. The anvil **10** is an output unit in the impact tool **1B** and is operable with a rotational force from the motor **6**.

(190) The anvil shaft **10C** in the anvil **10** is rotatably supported by an anvil bearing **146**. The anvil bearing **146** is an oilless bearing that is a slide bearing. The anvil bearing **146** is formed from iron. The anvil bearing **146** is cylindrical.

(191) The anvil bearing **146** in the embodiment is held by the front cylinder **4B** in the hammer case **104** with a bush **81**. The bush **81** is formed from iron. The bush **81** is fixed to the hammer case **104** by insert molding.

(192) The bush **81** includes a cylinder **81A** and a flange **81B**. The cylinder **81A** is located inward from the front cylinder **4B**. The flange **81B** extends outward in the radial direction from the rear end of the cylinder **81A**. The cylinder **81A** is fixed to the inner circumferential surface of the front cylinder **4B**. The flange **81B** is fixed to the rear surface of the annular portion **4C**. The front cylinder **4B** in the embodiment includes a protrusion **4N**. The protrusion **4N** protrudes inward in the radial direction from the inner circumferential surface of the front cylinder **4B**. The protrusion **4N** allows engagement to firmly fix the bush **81** and the front cylinder **4B** to each other.

(193) The bush **81** is insert-molded in the hammer case **104**. The anvil bearing **146** is then press-fitted into the cylinder **81A** in the bush **81**. As described above, the hammer case **104** is formed from aluminum, whereas the bush **81** is formed from iron. The bush **81** has a higher hardness than the front cylinder **4B**. The anvil bearing **146** is press-fitted into the bush **81** having a higher hardness. The anvil bearing **146** is thus fixed firmly to the bush **81**. The anvil bearing **146** with a smaller dimension in the front-rear direction can be fixed firmly to the bush **81**.

(194) The bush **81** may be fixed to the hammer case **104** with a method other than insert molding. The bush **81** may be fixed to the front cylinder **4B** after the surface of the bush **81** is knurled or serrated.

(195) The flange **81B** protects the rear surface of the annular portion **4C**. The anvil projections **10D** may have the front surfaces in contact with the rear surface of the annular portion **4C**. With the flange **81B** located on the rear surface of the annular portion **4C**, the front surfaces of the anvil projections **10D** hit the flange **81B** before hitting the rear surface of the annular portion **4C**. This reduces wear of the rear surface of the annular portion **4C**.

(196) The light unit **18** is located at the front of the hammer case **104**. The light unit **18** is annular. The light unit **18** surrounds the front cylinder **4B**. The light unit **18** surrounds the anvil shaft **10C** with the front cylinder **4B** in between.

(197) As in the above embodiment, the light unit **18** includes the COB light **50** and the optical member **57**. The optical member **57** is at least partially located frontward from the COB light **50**. The optical member **57** includes the outer cylinder **57A**, the inner cylinder **57B**, and the light transmitter **57C**. The outer cylinder **57A** is located radially outward from the COB light **50**. The inner cylinder **57B** is located radially inward from the COB light **50**. The light transmitter **57C** is located frontward from the COB light **50**. Light emitted from the COB light **50** passes through the light transmitter **57C** and illuminates an area ahead of the light unit **18**. The shock absorber **80** is located behind the COB light **50**.

(198) The projection **4D** on the hammer case **104** is located radially outward from the COB light **50**. The projection **4D** protrudes frontward from the COB light **50**. The projection **4D** is substantially annular. The projection **4D** is located radially outward from the optical member **57**. The light transmitter **57C** has its front surface located rearward from the front end face of the projection **4D**.

(199) The hammer case **104** has the groove **4E** radially inward from the projection **4D**. The groove **4E** is located on the front of the hammer case **104**. The groove **4E** is annular in a plane perpendicular to the rotation axis **AX**. The groove **4E** is recessed rearward from the front end of the hammer case **104**. The groove **4E** is between the front cylinder **4B** and the projection **4D** in the radial direction. The groove **4E** surrounds the anvil shaft **10C**. The light unit **18** is received in the groove **4E**.

(200) The circlip **116** is received in the circlip groove **4M** located on the front cylinder **4B**. The circlip groove **4M** is located frontward from the front surface of the light transmitter **57C**. The circlip **116** supports the optical member **57** from the front. The circlip **116** is in contact with the radially inner edge of the front surface of the light transmitter **57C**.

(201) The bumper **90** covers the front end of the surface of the hammer case **104**. The bumper **90** is annular. The bumper **90** is located outward in the radial direction from the light unit **18**. The bumper **90** protects the light unit **18**. The bumper **90** protects at least a part of the hammer case **104**.

(202) The light unit **18** is aligned with at least a part of the bumper **90** in the front-rear direction. More specifically, the bumper **90** overlaps the light unit **18** in the axial direction.

(203) The light unit **18** is aligned with at least a part of the anvil bearing **146** in the front-rear direction. More specifically, the anvil bearing **146** overlaps the light unit **18** in the axial direction.

(204) The bumper **90** is formed from an elastomer. The elastomer in the bumper **90** may be a thermoplastic elastomer or a thermosetting elastomer. The bumper **90** has a rubber hardness H_s (JIS A) of less than or equal to 100.

(205) The bumper **90** covers the outer circumferential surface and the front end face of the projection **4D**. The bumper **90** includes a cylindrical portion **91**, an annular portion **92**, and a protrusion **93**. The cylindrical portion **91** covers the outer circumferential surface of the projection **4D**. The annular portion **92** covers the front end face of the projection **4D**. The protrusion **93** protrudes inward in the radial direction from the inner circumferential surface of the cylindrical portion **91**. The protrusion **93** is annular in a plane perpendicular to the rotation axis **AX**.

(206) The projection **4D** has the bumper groove **4L** on its outer circumferential surface. The bumper groove **4L** surrounds the rotation axis **AX**. The protrusion **93** is received in the bumper groove **4L** on the outer circumferential surface of the projection **4D**.

(207) The bumper **90** has its rear end located frontward from the front end of the hammer **47** in the front-rear direction.

(208) As shown in FIG. **19**, a slit **105** is located in a lower portion of the front end of the hammer case **104**. The protrusion **57D** in the optical member **57** fits in the slit **105**. This restricts rotation of the light unit **18** relative to the hammer case **104**.

(209) As shown in FIGS. **19**, **20**, and **21**, the bumper **90** includes a projection **94** protruding downward from the lower portion of the cylindrical portion **91**. As shown in FIG. **21**, the projection

94 is held between the left housing **2L** and the right housing **2R** in the lateral direction. The cover **21C** in the motor compartment **21** covers the projection **94** from the front. This reduces the likelihood of the bumper **90** slipping frontward from the housing **2** or restricts rotation of the bumper **90** relative to the housing **2**.

(210) As described above, the elastomer bumper **90** surrounds the projection **4D** in the embodiment. The bumper **90** protects the light unit **18**. When, for example, the impact tool **1B** hits a surrounding object, the bumper **90** hits the object to reduce shock from the object to the light unit **18** with an elastic force from the bumper **90**. This reduces deterioration in the emission performance of the light unit **18**.

Modifications

(211) Modifications of the impact tool **1B** will now be described.

(212) In the above embodiment, the bumper **90** is located outward in the radial direction from the optical member **97**. The bumper **90** may have its front end facing the outer edge of the front surface of optical member **97**. The bumper **90** may serve as a retainer to restrict forward movement of the light unit **18** from the groove **4E**.

(213) In the above embodiment, the bumper **90** surrounds the projection **4D**. The bumper **90** has its rear end located frontward from the front end of the hammer **47**. The bumper **90** may have its rear end located rearward from the front end of the hammer **47**. The bumper **90** may cover the entire surface of the hammer case **104**.

(214) In the above embodiment, the bumper **90** is replaceable for the hammer case **104**. With multiple bumpers **90** in different colors, a bumper **90** in a first color may be attached to a first impact tool, and a bumper **90** in a second color may be attached to a second impact tool. The operator can differentiate multiple impact tools from one another by the colors of the bumpers **90**.

(215) In the above embodiment, the bumper **90** may be at least partially formed from a luminous material. This allows the operator to identify the impact tool **1B** at, for example, nighttime.

(216) FIG. **22** is a partially enlarged longitudinal sectional view of an impact tool **1B** according to a modification. As shown in FIG. **22**, the projection **4D** on the hammer case **104** may be eliminated. The bumper **90** may directly support the light unit **18** from radially outside. The bumper **90** may be in contact with the outer cylinder **57A** in the optical member **57** in the light unit **18**.

(217) As shown in FIG. **22**, the flange **81B** in the bush **81** may have a recess **81C** on its rear surface. The recess **81C** may contain a lubricant oil. The lubricant oil reduces sliding resistance at a contact between the front surfaces of the anvil projections **10D** and the rear surface of the flange **81B**.

(218) In the example in FIG. **22**, the bumper **90** may be integral with the optical member **57**.

(219) In the above embodiment, the bumper **90** may be fixed to the hammer case **104** with an adhesive. The bumper **90** may be fixed to the hammer case **104** with fasteners such as screws or rivets.

(220) In the above embodiment, the hammer case **104** may be integral with the bumper **90**.

(221) The bumper **90** may be fixed to the hammer case **104** by insert molding.

Other Embodiments

(222) In the above embodiments, the groove **4E** may be replaced with multiple nonannular grooves **4E** surrounding the anvil shaft **10C** at intervals. Each groove **4E** may receive a COB light chip and an optical member.

(223) In the above embodiments, the light **50** is a COB light. The light **50** may be any light including a substrate and a light emitter on the substrate, rather than a COB light.

(224) In the above embodiments, the impact tool **1** is an impact wrench.

(225) The impact tool **1** may be an impact driver.

(226) In the above embodiments, the impact tool **1** may use utility power (alternating current power supply) instead of the battery pack **25**. In the above embodiments, the striker **9** may be an oil pulse mechanism also called to as a fluid-drive, a stealth force, a hydraulic impact or a quiet strike.

REFERENCE SIGNS LIST

(227) **1** impact tool **1B** impact tool **2** housing **2A** upper end **2B** front end **2C** opening **2H** screw boss **2L** left housing **2R** right housing **2S** screw **3** rear cover **3S** screw **4** hammer case **4A** rear cylinder **4B** front cylinder **4C** annular portion **4D** projection **4E** groove **4F** snap ring groove **4H** screw boss **4Ha** screw boss **4Hb** screw boss **4J** facing surface **4K** protrusion **4L** bumper groove **4M** circlip groove **4N** protrusion **5** screw **6** motor **7** reducer **8** spindle **8A** flange **8B** spindle shaft **8C** ring portion **8D** spindle groove **8E** protrusion **8F** grease groove **9** striker **10** anvil (output unit) **10B** recess **10C** anvil shaft **10D** anvil projection **11** ball **12** fan **12A** bush **13** battery mount **14** trigger lever **15** forward-reverse switch lever **16** snap ring **17** seal **18** light unit **19** inlet **20** outlet **21** motor compartment **21C** cover **21D** upper surface **22** grip **23** battery holder **24** bearing box **25** battery pack **26** stator **27** rotor **28** stator core **29** front insulator **29S** screw **30** rear insulator **31** coil **32** rotor core **33** rotor shaft **34** rotor magnet **35** sensor magnet **37** sensor board **38** fusing terminal **39** rotor bearing **40** rotor bearing **41** pinion gear **42** planetary gear **42P** pin **43** internal gear **44** spindle bearing **45** washer **46** anvil bearing **46A** outer sleeve **46B** inner sleeve **46C** flange **47** hammer **47A** hammer groove **47B** hammer projection **47C** recess **47D** body **47E** rear outer cylinder **47F** front outer cylinder **47G** inner cylinder **47H** ball groove **48** ball **49** coil spring **49A** first coil spring **49B** second coil spring **50** chip-on-board light-emitting diode (COB LED) **51** substrate **51A** ring portion **51B** support **52** LED chip (light emitter) **54** bank **55** phosphor **57** optical member **57A** outer cylinder **57B** inner cylinder **57C** light transmitter **57D** protrusion **57E** incident surface **57F** emission surface **57G** inner circumferential protrusion **57H** stopper **57J** recess **80** shock absorber **81** bush **81A** cylinder **81B** flange **81C** recess **90** bumper **91** cylindrical portion **92** annular portion **93** protrusion **94** projection **104** hammer case **105** slit **116** circlip **146** anvil bearing **180** lead wire
AX rotation axis

Claims

1. An impact tool, comprising: a motor; a striker rotatable by the motor; an anvil with an anvil shaft and strikable by the striker in a rotation direction; a light that is annular, that surrounds the anvil shaft and that is configured to illuminate an area adjacent to a front end of the anvil; and a hammer case formed from metal, accommodating the striker and holding the light, the hammer case including a projection located radially outward from the light.
2. The impact tool according to claim 1, wherein the projection protrudes frontward from the light.
3. The impact tool according to claim 1, wherein the projection surrounds the anvil.
4. The impact tool according to claim 3, further comprising: an optical member including a light transmitter located frontward from the light, wherein the projection is located radially outward from the optical member.
5. The impact tool according to claim 4, wherein the anvil shaft is located radially inward from the optical member, the hammer case includes a front cylinder surrounding the anvil shaft, the impact tool further comprises a snap ring received in a snap ring groove on the front cylinder, and the snap ring supports the optical member from front.
6. The impact tool according to claim 4, wherein the optical member includes a protrusion located downward from the light transmitter, the impact tool further comprises a housing fastened to a rear of the hammer case and accommodating the motor, and the housing includes a cover covering the protrusion from front.
7. The impact tool according to claim 6, wherein the hammer case has a facing surface facing an upper surface of the cover, and the facing surface extends in a lateral direction.
8. The impact tool according to claim 4, wherein the anvil includes an anvil shaft located radially inward from the optical member, the impact tool further comprises an anvil bearing held by the hammer case and supporting the anvil shaft in a rotatable manner, and the striker has a front end located frontward from a rear end of the anvil bearing.

9. The impact tool according to claim 1, further comprising: a housing fastened to a rear of the hammer case with a screw, the housing accommodating the motor, wherein the hammer case includes a screw boss engaged with the screw, and at least a part of the housing covers the screw boss.
10. The impact tool according to claim 1, further comprising: a bumper that is annular and comprising an elastomer, the bumper being located outward in a radial direction from the light, wherein the light is located at a front of the hammer case.
11. The impact tool according to claim 10, wherein the hammer case has a groove that is annular on a front portion of the hammer case, and the light is received in the groove.
12. The impact tool according to claim 11, further comprising: an anvil bearing supporting the anvil in a rotatable manner, wherein the hammer case includes a rear cylinder accommodating the striker, a front cylinder located frontward from the rear cylinder, the front cylinder holding the anvil bearing, an annular portion connecting a front end of the rear cylinder and a rear end of the front cylinder, and a projection protruding frontward from an outer edge of the annular portion, the groove is between the front cylinder and the projection, and the bumper covers an outer circumferential surface of the projection.
13. The impact tool according to claim 12, wherein the bumper covers a front end face of the projection.
14. The impact tool according to claim 12, wherein the bumper includes a cylindrical portion covering the outer circumferential surface of the projection, and a protrusion protruding inward in the radial direction from an inner circumferential surface of the cylindrical portion, the projection has a bumper groove on the outer circumferential surface, and the protrusion is received in the bumper groove.
15. The impact tool according to claim 10, wherein the bumper has a rubber hardness Hs JIS A of less than or equal to 100.
16. The impact tool according to claim 10, wherein the striker includes a hammer rotatable by the motor, and the bumper is located frontward from the hammer in a front-rear direction.
17. The impact tool according to claim 1, wherein the light is a chip-on-board light.
18. The impact tool according to claim 1, wherein the light is in an annular groove and a radially outer extent of the annular groove is bounded by the projection.
19. The impact tool according to claim 18, wherein the projection is a portion of a unitary component that includes the annular groove.
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